



Complex Numbers : RACE-01

Subjective Type Question

1. Find the value of the sum $\sum_{n=1}^5 (i^n + i^{n+2})$. MCN231
2. Find the value of $13 \left(1 + \frac{2i}{3} + \left(\frac{2i}{3} \right)^2 + \left(\frac{2i}{3} \right)^3 + \dots \infty \right)$. MCN232
3. Express the set $A = \{i^n + i^{-n} : n \in \mathbb{Z}\}$ in roster form. MCN233
4. Find the absolute value of $\frac{i^{592} + i^{590} + i^{588} + i^{586} + i^{584}}{i^{582} + i^{580} + i^{578} + i^{576} + i^{574}}$. MCN234
5. Given that $x, y \in \mathbb{R}$. Solve the following equations and find (x, y) .
 (a) $(x + 2y) + i(2x - 3y) = 5 - 4i$ (b) $(x + iy) + (7 - 5i) = 9 + 4i$
 (c) $x^2 - y^2 - i(2x + y) = 2i$ (d) $\frac{x-3}{3+i} + \frac{y-3}{3-i} = i$ MCN235
6. If $x + iy = \frac{1}{1 + \cos \theta + i \sin \theta}$, then find the value of $\frac{2x-1+3y}{4x-2+5y}$. MCN236
7. If $x + iy = \left(\frac{a+ib}{a-ib} \right)^2 - \left(\frac{a-ib}{a+ib} \right)^2$, then find the value of $2x + y$; given that $x, y, a, b \in \mathbb{R}$. MCN237
8. If $a + ib = \frac{c+i}{c-i}$, where $a, b, c \in \mathbb{R}$, then find the value of $\frac{2ac(a^2 + b^2)}{b(c^2 - 1)}$. MCN238
9. Simplify the following complex numbers and express the result in the form of $a + bi$ where $a, b \in \mathbb{R}$
 (a) $\left(\frac{1+2i}{2+i} \right)^2$ (b) $-i(9+6i)(2-i)^{-1}$ (c) $\left(\frac{4i^3 - i}{2i+1} \right)^2$ (d) $\frac{(2+i)^2}{2-i} - \frac{(2-i)^2}{2+i}$ MCN239

10. Determine least positive integral value of n for which $\left(\frac{1+i}{1-i}\right)^n = 1$. MCN240
11. Find the square root(s) of following complex numbers :
 (a) $9 + 40i$ (b) $-11 - 60i$ (c) $50i$ (d) $-15 - 8i$ MCN241
12. Solve the following equations for x and express the solution in the form $a + ib$, $a, b \in \mathbb{R}$.
 (a) $ix^2 - 3x - 2i = 0$ (b) $2(1+i)x^2 - 4(2-i)x - 5 - 3i = 0$ MCN242
13. If $(x + iy)^{\frac{1}{3}} = a + bi$, then find $\frac{\frac{x}{a^2} + \frac{y}{b^2}}{a^2 - b^2}$; given that $x, y, a, b \in \mathbb{R}$. MCN243
14. If $x = 1 + 2i$, then find the value of $13x - x^3 - 7x^2 - 16$. MCN244
15. If $f(x) = x^4 + 9x^3 + 35x^2 - x + 4$ and $g(x) = x^4 - x^3 + x^2 + 3x - 5$, then find the value of $g(2 + 3i) - f(-5 + 4i)$ MCN245

ANSWER KEY

1. 0 2. $9 + 6i$ 3. $0, 2, -2$ 4. 1
5. (a) $(1, 2)$; (b) $(2, 9)$; (c) $(-2, 2)$ or $\left(-\frac{2}{3}, -\frac{2}{3}\right)$; (d) $(-2, 8)$ 6. $\frac{3}{5}$
7. $\frac{8ab(a^2 - b^2)}{(a^2 + b^2)^2}$ 8. 1 9. (a) $\frac{7}{25} + \frac{24}{25}i$; (b) $\frac{21}{5} - \frac{12}{5}i$; (c) $3 + 4i$; (d) $\frac{22}{5}i$
10. $n = 4$ 11. (a) $\pm(5 + 4i)$; (b) $\pm(5 - 6i)$; (c) $\pm 5(1 + i)$; (d) $\pm(1 + 4i)$
12. (a) $-i, -2i$ (b) $\frac{3-5i}{2}$ or $-\frac{1+i}{2}$ 13. 4 14. 29 15. $83 - 108i$



Complex Numbers : RACE-02

Subjective Type Question

1. If $z_1 = 1 + 2i$ and $z_2 = 3 + 5i$, and then find $\operatorname{Re}\left(\frac{\bar{z}_2 z_1}{z_2}\right)$. MCN246
2. If $z = \left(\frac{4i^3 - i}{2i + 1}\right)^2 + \frac{1}{1 - \cos\theta + 2i\sin\theta}$; $\theta \in \mathbb{R}$, then find the value of $2\left(\frac{z + \bar{z}}{2}\right) + 3\left(\frac{z - \bar{z}}{2i}\right)$. MCN247
3. If $z = \frac{2 + i}{x + 3 + iy}$; $x, y \in \mathbb{R}$ then find $\frac{z + \bar{z}}{2}, \frac{z - \bar{z}}{2}$ in terms of x, y . MCN248
4. If the complex number $\frac{\sin\left(\frac{x}{2}\right) + \cos\left(\frac{x}{2}\right) + i \tan(x)}{1 + 2i \sin\left(\frac{x}{2}\right)}$ is equal to its conjugate, then find the set of all possible values of x . MCN249
5. If $z^4 + 4 = (z - \alpha)(z - \bar{\alpha})(z - \beta)(z - \bar{\beta})$, $\operatorname{Re}(\alpha) > 0, \operatorname{Re}(\beta) < 0, \operatorname{Im}(\alpha) > 0, \operatorname{Im}(\beta) < 0$, then find α and β . MCN250
6. Prove that $z_1(\overline{z_2 + z_3}) + z_2(\overline{z_3 + z_1}) + z_3(\overline{z_1 + z_2}) = 2\operatorname{Re}(z_1\bar{z}_2 + z_2\bar{z}_3 + z_3\bar{z}_1) = 2\operatorname{Re}(\bar{z}_1 z_2 + \bar{z}_2 z_3 + \bar{z}_3 z_1)$ MCN251
7. If the equation, $z^2 - (3 + i)z + m + 2i = 0$; $m \in \mathbb{R}$ has a real root, then find m . MCN252
8. If a, b, c, a_1, b_1, c_1 , are non-zero complex numbers satisfying $\frac{a}{a_1} + \frac{b}{b_1} + \frac{c}{c_1} = 1 + i$ and $\frac{a_1}{a} + \frac{b_1}{b} + \frac{c_1}{c} = 0$, then prove that $\frac{a^2}{a_1^2} + \frac{b^2}{b_1^2} + \frac{c^2}{c_1^2} = 2i$. MCN253
9. Consider a quadratic equation $az^2 + bz + c = 0$ where a, b, c are complex numbers. Find the condition that the equation has one purely imaginary root. MCN254
10. Consider a quadratic equation $az^2 + bz + c = 0$ where a, b, c are complex numbers. Find the condition that the equation has two purely imaginary roots. MCN255

11. Given that the equation $z^2 + uz + v = 0$ where u, v are non-zero complex numbers has a real root. Then prove that $(u + \bar{u})(u - \bar{u})(v - \bar{v}) = 2(v - \bar{v})^2 + (u - \bar{u})^2(v + \bar{v})$. MCN256
12. Find the number of real $z \in \{x + iy : x, y \in \mathbb{R}, xy = 0\}$ satisfying $\bar{z}^3 - i \cdot \bar{z} - 1 = 0$ MCN257
13. If a, b, c are real numbers and z is a complex number such that $a^2 + b^2 + c^2 = 1$ and $b + ic = (1 + a)z$, then prove that $\frac{1 - i \bar{z}}{1 + i \bar{z}} = \frac{a - ib}{1 + c}$. MCN258
14. Let a, b, c are real numbers in the polynomial $P(z) = 2z^4 + az^3 + bz^2 + cz + 3$. If two roots of the equation $P(z) = 0$ are 2 and i , then find the value of ' a '. MCN259
15. Solve the equation $z^2 - (3 - 2i)z = (5i - 5)$ and find $z (\in \mathbb{C})$. MCN260

ANSWER KEY

1. $\frac{22}{17}$ 2. $\frac{92 + 54 \cos \theta - 6 \cot \frac{\theta}{2}}{5 + 3 \cos \theta}$ 3. $\left(\frac{2(x+3)+y}{(x+3)^2+y^2}, \frac{(x-2y+3)i}{(x+3)^2+y^2} \right)$
4. $2n\pi, n\pi + \frac{\pi}{4}; n \in \mathbb{Z}$
5. $z^4 + 4 = (z + 1 + i)(z + 1 - i)(z - 1 + i)(z - 1 - i); \alpha = 1 + i \text{ and } \beta = -1 - i$
7. 2 9. $(b\bar{c} + c\bar{b})(a\bar{b} + \bar{a}b) + (c\bar{a} - a\bar{c})^2 = 0$ 10. $\frac{a}{\bar{a}} = -\frac{b}{\bar{b}} = \frac{c}{\bar{c}}$
12. 0 14. $-\frac{11}{2}$ 15. $z = (2 + i) \text{ or } (1 - 3i)$



Complex Numbers : RACE-03

Subjective Type Question

1. If $(1-i)(1-2i)\dots(1-ni) = x - iy$; $x, y \in \mathbb{R}$ then prove that $x^2 + y^2 = \prod_{r=1}^n (r^2 + 1)$
MCN261
2. If $x + iy = \sqrt{\frac{a-ib}{c-id}}$, then prove that $(x^2 + y^2)^2 = \frac{a^2 + b^2}{c^2 + d^2}$; $a, b, c, d, x, y \in \mathbb{R}$.
MCN262
3. If $a_1, a_2, \dots, a_n$ are the real roots of the polynomial $x^n + p_1x^{n-1} + p_2x^{n-2} + \dots + p_{n-1}x + p_n$ ($p_1, p_2, \dots, p_n$ are real), then prove that
 $(1+a_1^2)(1+a_2^2)\dots(1+a_n^2) = (1-p_2+p_4-\dots)^2 + (p_1-p_3+p_5-\dots)^2$.
MCN263
4. If $z_1, z_2 \in \mathbb{C}$, $z_1^2 + z_2^2 \in \mathbb{R}$, $z_1(z_1^2 - 3z_2^2) = 2$ and $z_2(3z_1^2 - z_2^2) = 11$, then find $z_1^2 + z_2^2$.
MCN264
5. If 'x' and 'a' are real numbers, then prove that
 $(x^2 + a^2)^7 = (x^7 - 21x^5a^2 + 35x^3a^4 - 7xa^6)^2 + (7x^6a - 35x^4a^3 + 21x^2a^5 - a^7)^2$.
MCN265
6. Using complex numbers, prove that
 $(1^2 + 2^2)(3^2 + 4^2)(5^2 + 7^2) = 95^2 + 15^2 = 45^2 + 85^2 = 41^2 + 87^2 = 69^2 + 67^2$.
MCN266
7. If z_1, z_2, z_3 be three complex numbers such that $|z_1| = 1$, $|z_2| = 2$, $|z_3| = 3$ and $|z_1 + z_2 + z_3| = 1$, then find $|9z_1z_2 + 4z_1z_3 + z_2z_3|$.
MCN267
8. For any two complex numbers z_1, z_2 and any two real numbers a and b , prove that
 $|az_1 - bz_2|^2 + |bz_1 + az_2|^2 = (a^2 + b^2)(|z_1|^2 + |z_2|^2)$.
MCN268
9. Show using the properties of modulus that $|2\bar{z} + 5|(\sqrt{2} - i)| = \sqrt{3}|2z + 5|$, where $z \in \mathbb{C}$.
MCN269
10. If $\frac{a}{|z_1 - z_2|} = \frac{b}{|z_2 - z_3|} = \frac{c}{|z_3 - z_1|}$; $a, b, c \in \mathbb{R}$, and $z_1, z_2, z_3 \in \mathbb{C}$, then prove that
 $\frac{a^2}{z_1 - z_2} + \frac{b^2}{z_2 - z_3} + \frac{c^2}{z_3 - z_1} = 0$.
MCN270

11. Find the modulus, argument and the principal argument of $\frac{2+i}{4i+(1+i)^2}$ **MCN271**
12. If $z \in \mathbb{C}, |z| = 1, \arg(z) = \theta$, then find $\operatorname{Re}\left(\frac{z(1-\bar{z})}{\bar{z}(1+z)}\right)$ in terms of θ . **MCN272**
13. Let z, w be complex numbers such that $\bar{z} + i\bar{w} = 0$ and $\operatorname{amp}(zw) = \pi$, then find $\operatorname{amp}(z)$. (Here $\operatorname{amp}(z)$ denotes the principle value of argument.) **MCN273**
14. Let $\operatorname{amp}(z)$ denote the principle value of argument. If z and w are non zero complex numbers such that $|zw| = 1$ and $\operatorname{amp}(z) - \operatorname{amp}(w) = \frac{\pi}{2}$, then find $\bar{z}w$. **MCN274**
15. Let $\operatorname{amp}(z)$ denote the principle value of argument. If z and w are two non-zero complex numbers such that $|z| = |w|$ and $\operatorname{amp} z + \operatorname{amp} w = \pi$, then prove that $z = -\bar{w}$ **MCN275**

ANSWER KEY

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|-----|-----------------------------|-----|------------------|-----|--|
| 4. | 5 | 7. | 6 | 11. | Modulus = $\frac{\sqrt{5}}{6}$, Arg = $2k\pi - \tan^{-1}2 (k \in \mathbb{Z})$. Principal Arg = $-\tan^{-1}2$ |
| 12. | $-2\sin^2 \frac{\theta}{2}$ | 13. | $\frac{3\pi}{4}$ | 14. | $-i$ |



Complex Numbers : RACE-04

Subjective Type Question

1. Express the following complex numbers in polar form and exponential form :

(i) $-2 + 2i$

(ii) $-1 - \sqrt{3}i$

(iii) $\frac{(1+7i)}{(2-i)^2}$

(iv) $(1 - \cos\theta + i\sin\theta); \theta \in (0, \pi)$

MCN276

2. Express the following complex numbers in the polar form :

(i) $\frac{1+i}{1-i}$

(ii) $\frac{2+6\sqrt{3}i}{5+\sqrt{3}i}$

MCN277

3. Prove that $\frac{(\cos\theta - i\sin\theta)^4}{(\sin\theta + i\cos\theta)^5} = \sin\theta - i\cos\theta \quad \forall \theta \in \mathbb{R}$.

MCN278

4. Find the value of $\frac{4(i\cos 11^\circ - \sin 11^\circ)}{0.4(i\cos 34^\circ + \sin 34^\circ)}$ using Euler's formula.

MCN279

5. Let $\text{amp}(z)$ denote the principle value of argument. If z is a non-real complex number satisfying

$$|z| = 1, \text{ then prove that } z = \frac{1 + i \tan\left(\frac{\text{amp}(z)}{2}\right)}{1 - i \tan\left(\frac{\text{amp}(z)}{2}\right)}.$$

MCN280

6. If z is a complex number such that $|z| = r, \arg(z) = \theta$, then find $|e^{iz}|$ in terms of r and θ .

MCN281

7. If α, β are the roots of the equation $x^2 - 2x + 4 = 0$, then prove that $\alpha^n + \beta^n = 2^{n+1} \cos \frac{n\pi}{3}$ for all $n \in \mathbb{N}$.

MCN282

8. If $n \in \mathbb{N}$, then prove that $\left(\frac{1 + \sin\phi + i\cos\phi}{1 + \sin\phi - i\cos\phi}\right)^n = \cos\left(\frac{n\pi}{2} - n\phi\right) + i\sin\left(\frac{n\pi}{2} - n\phi\right) \quad \forall \phi \in \mathbb{R}$.

MCN283

9. Let z be a complex number with non-zero imaginary part and $|z| = 1$. Show that z can be represented as $\frac{c+i}{c-i}$ where c is real.

MCN284

10. If $z = \cos \frac{\pi}{8} + i \sin \frac{\pi}{8}$ and $A = \{z^n : n \in \mathbb{N}\}$, then find $n(A)$. MCN285
11. If 'a' is a complex number such that $|a| = 1$, find the values of a , so that equation $az^2 + z + 1 = 0$ has one purely imaginary root. MCN286
12. If z is a complex number such that $z + \frac{1}{z} = 2 \cos 3^\circ$, and $\omega = \left(\frac{\sqrt{3}}{2} - \frac{i}{2}\right)^{15}$, then find absolute value of $z^{2000} + \frac{1}{z^{2000}} + \text{Im}(\omega)$. MCN287
13. If $\cos \alpha + \cos \beta + \cos \gamma = 0$ and $\sin \alpha + \sin \beta + \sin \gamma = 0$, then prove the following :
 (a) $\sum \sin(\alpha + \beta) = 0 = \sum \cos(\alpha + \beta)$
 (b) $\sum \sin^2 \alpha = \frac{3}{2} = \sum \cos^2 \alpha$
 (c) $\cos(\alpha - \beta) + \cos(\beta - \gamma) + \cos(\gamma - \alpha) = \frac{-3}{2}$ MCN288
14. Prove that $e^{2mi\cot^{-1}p} \left(\frac{pi+1}{pi-1}\right)^m = \frac{(1-i)^{20}(\sqrt{3}+i)^{10}}{(-1-i\sqrt{3})^{20}}$; where $p > 0$ and $m \in \mathbb{Z}$. MCN289
15. If $z_r = \cos \frac{2r\pi}{5} + i \sin \frac{2r\pi}{5}$, $r = 0, 1, 3, 4, \dots$, then find $z_1 z_2 z_3 z_4 z_5$. MCN290

ANSWER KEY

1. (i) $2\sqrt{2}\left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4}\right); 2\sqrt{2}e^{i\left(\frac{3\pi}{4}\right)}$; (ii) $2\left(\cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3}\right); 2e^{i\left(\frac{4\pi}{3}\right)}$;
 (iii) $\sqrt{2}\left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4}\right); \sqrt{2}e^{i\left(\frac{3\pi}{4}\right)}$; (iv) $2\sin\left(\frac{\theta}{2}\right)\left(\cos\left(\frac{\pi}{2} - \frac{\theta}{2}\right) + i \sin\left(\frac{\pi}{2} - \frac{\theta}{2}\right)\right); 2\sin\left(\frac{\theta}{2}\right)e^{i\left(\frac{\pi}{2} - \frac{\theta}{2}\right)}$
2. (i) $z = \cos \frac{\pi}{2} + i \sin \frac{\pi}{2}$ (ii) $z = 2\left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}\right)$
4. $\frac{10}{\sqrt{2}}(1+i)$ 6. $e^{-rsin\theta}$ 10. 16
11. $a = \cos \theta + i \sin \theta$, where $\theta = \cos^{-1}\left(\frac{\sqrt{5}-1}{2}\right)$ 12. 2 15. 1



Complex Numbers : RACE-05

Subjective Type Question

- Locate the points representing the complex number z on the Argand plane satisfying the following equations :
 - $|z + 1 - 2i| = \sqrt{7}$
 - $|z - 1|^2 + |z + 1|^2 = 4$
 - $\left| \frac{z-3}{z+3} \right| = 3$
 - $|z - 3| = |z - 5 + 6i|$

MCN291
- Locate the points representing the complex number z on the Argand plane satisfying the following equations :
 - $|z - 11 - 2i| < 5$
 - $|z - 12i - 10| > 10$
 - $10 < |z - 12 + 10i| < 20$
 - $5 \leq |z - 2 - 3i| \leq 7$

MCN292
- Find center and radius of circle represented by $z\bar{z} + (2 - 3i)z + (2 + 3i)\bar{z} + 4 = 0$

MCN293
- If z_1, z_2, z_3, z_4 are the affixes of four points in the Argand plane and z is the affix of a point such that $|z - z_1| = |z - z_2| = |z - z_3| = |z - z_4|$, then prove that z_1, z_2, z_3, z_4 are concyclic,

MCN294
- Locate the points representing the complex number z on the Argand plane satisfying the following equations :
 - $|z - 1| + |z + 1| = 2$
 - $|z - i| + |z + 1| = 2$
 - $|z - 12i| + |z + 5| < 20$
 - $|z - 12| + |z + 5i| = 10$

MCN295
- Locate the points representing the complex number z on the Argand plane satisfying the following equations :
 - $\|z - i\| - \|z + 2i\| = 3$
 - $\|z - 4i\| - \|z + 3\| = 2$
 - $|z - 12i| - |z + 5| = 20$
 - $|z - 8i| - |z + 6| = 5$

MCN296
- Find range of k , if the points z satisfying $|z + i| - |z - i| = k$ lie on a hyperbola.

MCN297
- If locus of z satisfying $|z - i| + |z + 2i| = k, k > 0$, is an ellipse, then find range of k^2 .

MCN298
- Prove that the locus of $P(z)$ satisfying $\left| \frac{z - z_1}{z - z_2} \right| = k$ ($k \neq 1, > 0$) is a circle whose center divides $A(z_1)$ and $B(z_2)$ in ratio $k^2 : 1$ externally.

MCN299
- Let z_1, z_2 are complex numbers satisfying $|z_1| = 12$ and $|z_2 - 3 - 4i| = 5$. Find the minimum value of $|z_1 - z_2|$.

MCN300

11. If the complex number z satisfies $|z| \geq 3$, then find the least value of $\left|z + \frac{1}{z}\right|$. MCN301
12. If z is any complex number satisfying $|z - 3 - 2i| \leq 2$, then find the minimum value of $|2z - 6 + 5i|$. MCN302
13. Find the number of solutions of the equations $|z - (4 + 8i)| = \sqrt{10}$ and $|z - (3 + 5i)| + |z - (5 + 11i)| = 4\sqrt{5}$. MCN303
14. Let $|z| = 2$ and $w = \frac{z+1}{z-1}$ where $z, w \in \mathbb{C}$. If M and m respectively be the greatest and least modulus of w , then find the value of $(2010m + M)$. MCN304
15. If r_1 and r_2 are the maximum and minimum distances of a point on the curve $10(z\bar{z}) - 3i\{z^2 - (\bar{z})^2\} - 6 = 0$ from origin, then find $r_1 + r_2$. MCN305

ANSWER KEY

1. (a) Circle of radius $\sqrt{7}$ with centre $(-1, 2)$
 (b) Unit circle with centre at origin
 (c) Circle with centre $\left(-\frac{15}{4}, 0\right)$ and radius $\left(\frac{9}{4}\right)$
 (d) Straight line $x - 3y = 13$
2. (a) Inside a circle with center $(11, 2)$ and radius 5.
 (b) Outside a circle with center $(10, 12)$ and radius 10.
 (c) Between two concentric circles with center $(12, -10)$ and radii 10, 20.
 (d) On or between two concentric circles with center $(2, 3)$ and radii 5, 7.
3. Center = $(-2, -3)$, Radius = 3
5. (a) Line segment AB , where $A(1, 0)$ and $B(-1, 0)$
 (b) Ellipse with eccentricity $\frac{1}{\sqrt{2}}$ and foci $A(0, 1), B(-1, 0)$
 (c) Inside an ellipse with eccentricity $\frac{13}{20}$ and foci $A(0, 12), B(-5, 0)$
 (d) No such point exists
6. (a) Points on line AB , where $A(1, 0)$ and $B(-1, 0)$ excluding those between A and B .
 (b) Hyperbola with eccentricity $\frac{5}{2}$ and foci $A(0, 4), B(-3, 0)$
 (c) No such point exists
 (d) Branch of Hyperbola with eccentricity 2 and foci $A(0, 8), B(-6, 0)$ closer to B .
7. $-2 < k < 2, k \neq 0$ 8. $9, \infty$ 10. 2 11. $\frac{8}{3}$
12. 5 13. 2 14. 673 15. 3



Complex Numbers : RACE-06

Single Correct Question

1. If z is a complex number, then locus of z satisfying $z^2 + \bar{z}^2 = 2$ is
 (A) a circle (B) a straight line (C) a hyperbola (D) an ellipse

MCN306

Subjective Type Question

2. Find the number of solutions of $\operatorname{Re}(z^2) = 0$ and $|z| = a\sqrt{2}$, where z is a complex number and $a > 0$.

MCN307

3. Let z be a complex number such that $z = (1-t) + i\sqrt{t^2+t+2}$ where t is a real parameter. Then prove that the locus of z on Argand Plane is a hyperbola.

MCN308

4. If $z \neq 1$ and $\frac{z^2}{z-1}$ is real, then prove that the point represented by the complex number z lies either on the real axis or on a circle passing through the origin

MCN309

5. Prove that $\left\{ \operatorname{Re}\left(\frac{2iz}{1-z^2}\right) : z \text{ is a complex number, } |z| = 1, z \neq \pm 1 \right\} = (-\infty, -1] \cup [1, \infty)$.

MCN310

6. Find all complex numbers z satisfying the equation $2|z|^2 + z^2 - 5 + i\sqrt{3} = 0$.

MCN311

7. Let $z = x + iy$ be a complex number where x and y are integers. Then find the area of the rectangle whose vertices are the roots of the equation $z\bar{z}^3 + \bar{z}z^3 = 350$.

MCN312

8. Let z be a complex number such that $z \in \mathbb{C} \setminus \mathbb{R}$ and $\frac{1+z+z^2}{1-z+z^2} \in \mathbb{R}$, then prove that $|z| = 1$.

MCN313

9. Find the minimum value of the expression $E = |z|^2 + |z-3|^2 + |z-6i|^2$
 (where $z = x + iy, x, y \in \mathbb{R}$)

MCN314

10. If z_1 and z_2 are two given complex numbers, then prove that the locus of z satisfying $|z-z_1|^2 + |z-z_2|^2 = k$, is a circle having centre at $\frac{z_1+z_2}{2}$ and radius $= \frac{1}{2}\sqrt{2k - |z_1-z_2|^2}$.

MCN315

11. If A, B, C & D are four points representing the complex numbers z_1, z_2, z_3 & z_4 then prove that $AB \parallel CD$ if $\frac{z_4 - z_3}{z_2 - z_1}$ is purely real. MCN316
12. If A, B, C & D are four points representing the complex numbers z_1, z_2, z_3 & z_4 then prove that $AB \perp CD$ if $\frac{z_4 - z_3}{z_2 - z_1}$ is purely imaginary. MCN317
13. If $\bar{\alpha}z + \alpha\bar{z} + r = 0$, where r is real and α is non zero complex constant represents the equation of perpendicular bisector of the line segment joining $A(z_1)$ and $B(z_2)$, then prove that $\frac{\bar{\alpha}}{\bar{z}_1 - \bar{z}_2} = \frac{\alpha}{z_1 - z_2} = \frac{-r}{|z_1|^2 - |z_2|^2}$. MCN318
14. Prove that the perpendicular distance of the point z_c from the line $z\bar{a} + \bar{z}a + b = 0$, $(z, a \in \mathbb{C}, b \in \mathbb{R})$ is $\frac{|z_c\bar{a} + \bar{z}_ca + b|}{2|a|}$. MCN319
15. If ω denotes the orthocentre of $\triangle ABC$ where $A(z_1), B(z_2), C(z_3)$, then prove that $\frac{\omega - z_1}{\bar{\omega} - \bar{z}_1} + \frac{\omega - z_2}{\bar{\omega} - \bar{z}_2} + \frac{\omega - z_3}{\bar{\omega} - \bar{z}_3} + \frac{z_2 - z_1}{\bar{z}_2 - \bar{z}_1} + \frac{z_2 - z_3}{\bar{z}_2 - \bar{z}_3} + \frac{z_3 - z_1}{\bar{z}_3 - \bar{z}_1} = 0$ MCN320

ANSWER KEY

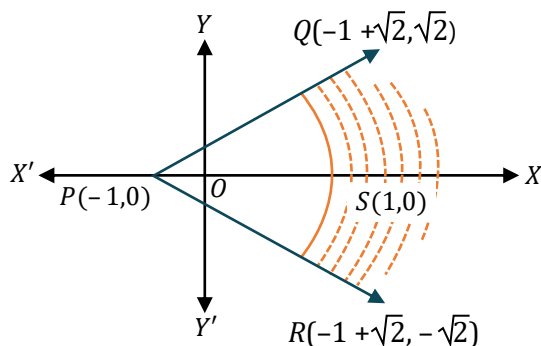
1. C 2. 4 6. $\frac{1}{\sqrt{6}} - \frac{3}{\sqrt{2}}i, -\frac{1}{\sqrt{6}} + \frac{3}{\sqrt{2}}i, \frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}}i, -\frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}}i$
7. 48 9. 30



Complex Numbers : RACE-07

Single Correct Question

1. The shaded region, where $P \equiv (-1, 0)$, $Q \equiv (-1 + \sqrt{2}, \sqrt{2})$, $R \equiv (-1 + \sqrt{2}, -\sqrt{2})$ and $S \equiv (1, 0)$ is represented by : (Here $\text{amp}(z)$ denotes the principle value of argument.)



- (A) $|z + 1| > 2$, $|\text{amp}(z + 1)| < \frac{\pi}{4}$ (B) $|z + 1| < 2$, $|\text{amp}(z + 1)| < \frac{\pi}{2}$
 (C) $|z - 1| > 2$, $|\text{amp}(z + 1)| > \frac{\pi}{4}$ (D) $|z - 1| < 2$, $|\text{amp}(z + 1)| > \frac{\pi}{2}$

MCN321

Subjective Type Question

2. Let $\text{amp}(z)$ denote the principle value of argument. Sketch the locus of z satisfying the following equations:

(i) $\text{amp}(z) = -\frac{5\pi}{6}$

(ii) $\text{amp}(z + i) = -\frac{3\pi}{4}$

(iii) $\text{amp}(z + 3 + 4i) = \frac{\pi}{2}$

(iv) $\text{amp}(z - 7 - 4i) = \pi$ (v) $\text{amp}(z + 7) = \frac{3\pi}{4}$

MCN322

3. Let $\text{amp}(z)$ denote the principle value of argument. If $|z - 25i| \leq 15$, then find $|\max(\text{amp}z) - \min(\text{amp}z)|$.

MCN323

4. Let $\arg(z)$ denote the principle value of argument. Prove that points z satisfying the equation $|z - 2 - i| = |z| \cdot \left| \sin\left(\frac{\pi}{4} - \arg(z)\right) \right|$ lie on a parabola.

MCN324

5. Find the area of the region of the argand plane described by complex numbers z satisfying $\frac{\pi}{6} < \text{amp}(z) < \frac{2\pi}{3}$ & $3 < |z| < 5$. Here $\text{amp}(z)$ denotes the principle value of argument.

MCN325

6. For any two complex numbers z_1, z_2 prove the following (Here $\arg(z)$ denotes the principle value of argument.)
- (i) $|z_1 + z_2|^2 = |z_1|^2 + |z_2|^2 + 2|z_1||z_2| \cos(\theta_1 - \theta_2)$ (θ_1, θ_2 are arguments of z_1, z_2)
- (ii) $|z_1 - z_2|^2 = |z_1|^2 + |z_2|^2 - 2|z_1||z_2| \cos(\theta_1 - \theta_2)$ (θ_1, θ_2 are arguments of z_1, z_2)
- (iii) $|z_1 + z_2| = |z_1 - z_2| \Leftrightarrow \arg(z_1) - \arg(z_2) = \pm \frac{\pi}{2} \Leftrightarrow \frac{z_1}{z_2}$ is purely imaginary ($z_2 \neq 0$)
- (iv) $|z_1 + z_2| = |z_1| + |z_2| \Leftrightarrow \arg(z_1) = \arg(z_2) \Leftrightarrow \frac{z_1}{z_2}$ is purely imaginary ($z_2 \neq 0$)
- (v) $|z_1 + z_2|^2 = |z_1|^2 + |z_2|^2 \Leftrightarrow \frac{z_1}{z_2}$ is purely imaginary ($z_2 \neq 0$)
- MCN326**
7. Among the complex numbers z satisfying the condition $|z + 3 - \sqrt{3}i| = \sqrt{3}$, find the number having the least positive principle argument.
- MCN327**
8. A complex number $z = 3 + 4i$ is rotated about another fixed complex number $z_1 = 1 + 2i$ in anticlockwise direction by 45° angle. Find the complex number represented by new position of z in argand plane.
- MCN328**
9. Let A, B, C are three points in argand plane (in anticlockwise sense) representing the complex numbers z_1, z_2, z_3 . Find conditions on ℓ, m, n , if $\frac{\ell z_1 + m z_2 + n z_3}{\ell + m + n}$ lies outside $\triangle ABC$, but inside $\angle ABC$.
- MCN329**
10. If $A(z_1), B(z_2), C(z_3)$ are the vertices of an equilateral $\triangle ABC$, taken anticlockwise then find $\arg\left(\frac{z_2 + z_3 - 2z_1}{z_3 - z_2}\right)$
- MCN330**
11. Find the center of the arc represented by locus of points z satisfying $\arg\left(\frac{z - 3i}{z - 2i + 4}\right) = \frac{\pi}{4}$
- MCN331**
12. If the area of the triangle formed by the points represented by $z, z + iz$ and iz is 200, then find the value of $|z|$.
- MCN332**
13. If the complex numbers z_1, z_2 and the z_3 form an equilateral triangle, then prove the following
- (a) $\frac{1}{z_1 - z_2} + \frac{1}{z_2 - z_3} + \frac{1}{z_3 - z_1} = 0$
- (b) $z_1^2 + z_2^2 + z_3^2 = z_1 z_2 + z_2 z_3 + z_3 z_1$
- (c) $z_1^3 + z_2^3 + z_3^3 = 3z_1 z_2 z_3$
- MCN333**

14. Consider the complex numbers $z_1 = 10 + 6i$ and $z_2 = 4 + 2i$. If $\arg\left(\frac{z - z_1}{z - z_2}\right) = \frac{\pi}{4}$, find the centre and radius of the locus traced by the complex number z .

MCN334

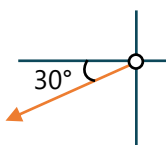
15. Find all complex numbers common to $|z| = 2\sqrt{2}$ and $\arg\left\{\frac{z - (5 + i)}{z - (1 + i)}\right\} = \frac{\pi}{4}$.

MCN335

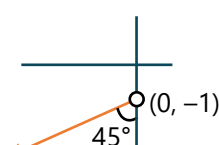
ANSWER KEY

1. A

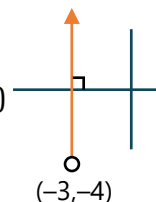
2. (i)



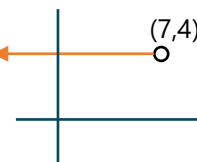
- (ii)



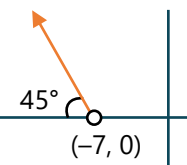
- (iii)



- (iv)



- (v)



3. $2\cos^{-1}\left(\frac{4}{5}\right)$

5. 4π

7. $3e^{i\frac{2\pi}{3}}$

8. $1 + (2 + 2\sqrt{2})i$

9. $(\ell < 0, m > 0, n > 0)$

10. $\frac{\pi}{2}$

11. $\frac{1}{2}(9i - 5)$

12. 20

14. $5 + 7i, \sqrt{26}$

15. $z = \left(\frac{3 - \sqrt{7}}{2}\right) + i\left(\frac{3 + \sqrt{7}}{2}\right)$



Complex Numbers : RACE-08

Subjective Type Question

1. Solve $z^3 = 8i$ and express its solutions in exponential form. MCN336

2. Solve $\left(\frac{z}{2}\right) + \left(\frac{z}{2}\right)^2 + \left(\frac{z}{2}\right)^3 + \left(\frac{z}{2}\right)^4 + \left(\frac{z}{2}\right)^5 = -1$. MCN337

3. Solve the equation $(z-2)^{11} = \frac{\sqrt{3}-1+i(\sqrt{3}+1)}{2\sqrt{2}}$. MCN338

4. If $\cos\theta + i\sin\theta$ is a root of the equation $x^n + a_1x^{n-1} + a_2x^{n-2} + \dots + a_{n-1}x + a_n = 0$ then find the absolute value of $\sum_{r=1}^n a_r \cos r\theta$ (where all coefficient are real). MCN339

5. Find the sum of the series $1(2-\omega)(2-\omega^2) + 2(3-\omega)(3-\omega^2) \dots (n-1)(n-\omega)(n-\omega^2)$ where ω is one of the non real cube roots of unity. MCN340

6. If ω is a non-real cube root of unity, then prove the following :
 (a) $x^3 + y^3 = (x+y)(x+y\omega)(x+y\omega^2)$
 (b) $x^3 + y^3 + z^3 - 3xyz = (x+y+z)(x+y\omega+z\omega^2)(x+y\omega^2+z\omega)$
 (c) $|a+b\omega+c\omega^2| = \sqrt{a^2+b^2+c^2-ab-bc-ca} = |a+b\omega^2+c\omega|$; $a, b, c \in \mathbb{R}$. MCN341

7. If $z^2 + z + 1 = 0$, where z is complex number, then find the value of $\left(z + \frac{1}{z}\right) + \left(z^2 + \frac{1}{z^2}\right)^2 + \left(z^3 + \frac{1}{z^3}\right) + \dots + \left(z^6 + \frac{1}{z^6}\right)^2$. MCN342

8. If l, ω, ω^2 are the cube roots of unity, then prove the following :
 (a) $(1-\omega+\omega^2)^5 + (1+\omega-\omega^2)^5 = 32$
 (b) $(1+\omega)(1+\omega^2)(1+\omega^4)(1+\omega^5) = 1$
 (c) $\frac{a+b\omega+c\omega^2}{a\omega+b\omega^2+c} = \omega^2$; $a, b, c \in \mathbb{R}$ MCN343

9. If a, b, c, d, e, f, g are the seventh roots of unity and $g \in \mathbb{R}$, then find the value of
 (a) $a^2 + b^2 + c^2 + d^2 + e^2 + f^2 + g^2$
 (b) $a^3 + b^3 + c^3 + d^3 + e^3 + f^3$
 (c) $a^{-5} + b^{-5} + c^{-5} + d^{-5} + e^{-5} + f^{-5}$
 (d) $a^7 + b^7 + c^7 + d^7 + e^7 + f^7$
 (e) $(1+a)(1+b)(1+c)(1+d)(1+e)(1+f)(1+g)$ MCN344
10. If the equation $(z+1)^7 + z^7 = 0$ has roots $z_1, z_2, \dots, z_7$, then find the value of
 (a) $\sum_{r=1}^7 \operatorname{Re}(z_r)$ and (b) $\sum_{r=1}^7 \operatorname{Im}(z_r)$ MCN345
11. Prove that $\cos \frac{\pi}{7} \cdot \cos \frac{2\pi}{7} \cdot \cos \frac{3\pi}{7} = \frac{1}{8}$ using 7th roots of unity. MCN346
12. If $1, \alpha_1, \alpha_2, \dots, \alpha_{2008}$ are 2009th roots of unity, then find the absolute value of
 $\sum_{r=1}^{2008} r(\alpha_r + \alpha_{2009-r})$ MCN347
13. If $Z_r; r = 1, 2, 3, \dots, 50$ are the roots of the equation $\sum_{r=0}^{50} (Z)^r = 0$, then find the value of $\left| \sum_{r=1}^{50} \frac{1}{Z_r - 1} \right|$ MCN348
14. If $a_i; i = 1, 2, 3, \dots, 11$ are the eleventh roots of unity, then find $\sum_{i=1}^{10} \left(\sum_{j=i+1}^{11} (a_i a_j) \right)$ MCN349
15. Evaluate $\sum_{p=1}^{32} (3p+2) \left(\sum_{q=1}^{10} \left(\sin \frac{2q\pi}{11} - i \cos \frac{2q\pi}{11} \right) \right)^p$ MCN350

ANSWER KEY

1. $2 \cdot e^{-i\frac{\pi}{2}}, 2 \cdot e^{i\frac{\pi}{6}}, 2 \cdot e^{i\frac{5\pi}{6}}$ 2. $2 \cdot e^{i\frac{k\pi}{3}}; k = 1, 2, 3, 4, 5$ 3. $2 + e^{\frac{i(24n+5)\pi}{132}}; n = 0, 1, 2, \dots, 10$
4. 1 5. $\left[\frac{n(n+1)}{2} \right]^2 - n$ 7. 6
9. (a) 0 (b) -1 (c) -1 (d) 6 (d) 2
10. (a) $-\frac{7}{2}$, (b) zero
12. 2009 13. 25 14. 0 15. $48(1-i)$